

Experiment 2

Name: Urvi Shivalkar

PRN: B25ET1196

Class: SY-3(C)

Title:

Code:

```
#include <stdio.h>
```

```
void insert(int a[], int n)
```

```
{
    int i;
    printf("Enter the array elements: ");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
}
```

```
void display(int a[], int n)
```

```
{
    int i;
    printf("Array elements are: ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    printf("\n");
}
```

```
void search(int a[], int n)
```

```
{
    int i, key, found = 0;

    printf("Enter the number to search: ");
    scanf("%d", &key);
    for(i = 0; i < n; i++)
    {
        if(a[i] == key)
        {
```

```

        printf("Number is found\n");
        found = 1;
        break;
    }
}
if(found == 0)
{
    printf("Number not found\n");
}
}
void binarySearch(int a[], int low, int high, int key)
{
    int mid;

    if(low > high)
    {
        printf("Number not found\n");
        return;
    }
    mid = (low + high) / 2;
    if(a[mid] == key)
    {
        printf("Number is found\n");
    }
    else if(key < a[mid])
    {
        binarySearch(a, low, mid - 1, key);
    }
    else
    {
        binarySearch(a, mid + 1, high, key);
    }
}

int main()
{
    int a[10], n, choice;

```

```

printf("Enter the no. of elements: ");
scanf("%d", &n);
insert(a, n);
do
{
    printf("\n----- MENU ----- \n");
    printf("1. Display\n");
    printf("2. Search\n");
    printf("3. BinarySearch\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            display(a, n);
            break;

        case 2:
            search(a, n);
            break;

        case 3:
            {
                int key;

                printf("Enter number for search: ");
                scanf("%d", &key);

                binarySearch(a, 0, n - 1, key);
                break;
            }

        case 4:

```

```

        printf("Exit\n");
        break;

    default:
        printf("Invalid choice\n");
    }

} while(choice != 4);

return 0;
}

```

Output:

```

Enter the no. of elements: 5
Enter the array elements: 11 22 33 44 55

----- MENU -----
1. Display
2. Search
3. BinarySearch
4. Exit
Enter your choice: 1
Array elements are: 11 22 33 44 55

```

```

Enter the no. of elements: 5
Enter the array elements: 11 22 33 44 55

----- MENU -----
1. Display
2. Search
3. BinarySearch
4. Exit
Enter your choice: 2
Enter the number to search: 22
Number is found

```

```

Enter the no. of elements: 5
Enter the array elements: 11 22 33 44 55

----- MENU -----
1. Display
2. Search
3. BinarySearch
4. Exit
Enter your choice: 3
Enter number for search: 55
Number is found

```

```

Enter the no. of elements: 5
Enter the array elements: 11 22 33 44 55

----- MENU -----
1. Display
2. Search
3. BinarySearch
4. Exit
Enter your choice: 4
Exit

```